

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: ***it seems that a malicious actor is performing a SYN flood attack***

The logs show that: ***almost 50 TCP SYN requests are coming from an unfamiliar IP address within about 3 seconds***

This event could be: ***a SYN flood attack a type of DoS attack***

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- 1. The employee website visitor is trying to access the webpage. Their browser is sending SYN requests to access the web page hosted on the server. A SYN packet is sent from the source to the destination.***
- 2. The web server responds with an acknowledgement (SYN/ACK) agreeing to the synchronization request.***
- 3. The visitor's computer sends an ACK packet acknowledging the permission to connect.***

Explain what happens when a malicious actor sends a large number of SYN packets all at once: ***It floods the system and uses up too many resources. The system responds by shutting down.***

Explain what the logs indicate and how that affects the server: ***The logs indicate a SYN flood attack. It will either shut the system down or cause interactions to be extremely slow.***